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Anti-backspin component for vascular prosthesis delivery device

Abstract

A delivery device prevents longitudinal expansion of an aortic prosthesis under compression by engagement of a clutch with a pin upon release of a handle rotating a gearing mechanism about the pin that advances the aortic prosthesis during endovascular implantation at a surgical site.

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Background/Summary

RELATED APPLICATION (1) This application claims the benefit of priority to U.S. Provisional Patent Application No. 63/043,197, filed Jun. 24, 2020, the entire teachings of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

(1) An aortic aneurysm is an enlargement or bulge in a section of the aorta, which can be life-threatening. Treatment of aortic aneurysms remain a challenge. Endovascular repair has become a viable alternative to open repair of an aortic aneurysm. An endovascular approach results in insertion of an endovascular graft to exclude the aneurysm sac from blood flow. Once in place, the endovascular graft is expanded to create a new path for blood flow. The endovascular graft remains inside the aorta permanently through the use of a metal stent creating a tight fit and seal against the wall of the aorta. Currently, endovascular delivery devices have limitations on the precise control that the physician has in placement of the graft at the site of the aneurysm.

(2) More specifically, endovascular implantation typically relies on delivery devices that employ different combinations of translational forces and mechanical advantage to negotiate vasculature and then to precisely target and deploy endovascular prostheses at a surgical site. During the course of delivery and deployment, endovascular prostheses are exposed to various physical forces that often distort the position of the prostheses within the delivery device. Typically, a primary force that affects disposition of a prosthesis within an endovascular delivery device is longitudinal compression consequent to advancement of the prosthesis from a sheath that radially confines the

prostheses until the prosthesis reaches the surgical site where it is to be deployed. Often, advancement is obtained by rotation of a handle by the surgeon about a body of the delivery device. The rotation is translated to longitudinal force along the body of the delivery device by gearing which advances the prosthesis within vasculature of the patient to the surgical site. Once the force advancing the prosthesis stops, the prosthesis will have a tendency to resume its original longitudinal dimension as an opposing force in a direction opposite to that of advancement of the prosthesis (i.e., in a proximal direction back toward the physician). This opposing force will be translated back through the gearing of the delivery device to cause the handle to “backspin” in a direction of rotation opposite to that the surgeon employed to advance the prosthesis when holding the handle. This can present at least three problems for the physician during implantation of the prosthesis. The first is an inability to precisely know the position of the prosthesis at the surgical site. Despite fluoroscopic imagery, if the prosthesis changes position as a consequence of even a momentary release of a handle employed by the surgeon to advance the prosthesis to the surgical site, the surgeon will be uncertain as to how much he needs to advance the handle to continue advancement of the prosthesis, because the prosthesis will reacquire its longitudinally compressed position before advancement of the prosthesis as a whole continues. The second problem is related to the first, and consists of a lack of sensitivity in overall control of the prosthesis during delivery and, most importantly, when landing the prosthesis during deployment and release of the prosthesis from the delivery device, which are portions of the procedure that, generally, are irreversible. The third is the possibility, during advancement of the prosthesis by rotation of the handle, that teeth of gears in the gearing assembly that translates rotation of the handle to longitudinal force can become jammed, freezing the delivery device until the surgeon can momentarily loosen the mesh between the gears, such as by some method that is not part of the method of delivery of the prosthesis. Jamming of the delivery device, and methods, sometimes ad hoc methods, such as shaking the delivery device during implantation of an arterial prosthesis, can be a distraction to the surgeon and the patient, who often is conscious during the procedure, and can endanger the success of the implantation of the prosthesis and the life of the patient.

(3) As a consequence, there is a need to develop new and improved delivery devices to treat aortic aneurysms.

SUMMARY OF THE INVENTION

(4) The invention is generally directed to a delivery device for implanting a vascular prosthesis that effectively prevents backspin of prostheses under longitudinal compression within the delivery device caused by advancement of the prosthesis to a surgical site.

(5) In one embodiment, a delivery device of the invention includes a handle body having a longitudinal axis, a proximal end, and a distal end. A gear rack extends within the handle body and a proximal handle extends about the gear rack and defines teeth. The proximal handle is rotatable about the handle body and the gear rack. A distal handle extends around handle body at the distal end of the handle body, and a guidewire catheter having a proximal end and a distal end extends through the handle body, the proximal handle, and the distal handle along the longitudinal axis of the handle body. A delivery catheter is axially fixed to the proximal handle and has a distal end extending from within the distal end of the handle body and about the guidewire catheter. An outer catheter extends distally from the distal handle and about the delivery catheter when the delivery catheter is in a first, retracted position. A gear assembly links the teeth of the proximal handle to the gear rack, whereby rotation of the proximal handle about the longitudinal axis moves the proximal handle and the delivery catheter along the longitudinal axis relative to the gear rack. A clutch at the gear assembly engages the gear assembly with the proximal handle and thereby biases longitudinal movement of the gear assembly and associated rotation of the proximal handle about the longitudinal axis.

(6) In one particular embodiment, the gear assembly of the delivery device includes a pinion gear assembly that includes an upper pinion gear engaged with the proximal handle, wherein the upper

pinion gear defines a non-circular pinion gear orifice that is rotatable about the pinion gear axis. A lower pinion gear in this embodiment is axially aligned with the upper pinion gear and defines a lower pinion gear orifice. The lower pinion gear is engaged with the gear rack and is selectively engaged with the upper pinion gear, wherein the clutch engages lower pinion gear when the gear assembly is directed in a proximal direction. In one particular example of this embodiment, the lower pinion gear includes a lower portion extending toward the longitudinal axis of the handle body, a gear portion engaged with the gear rack, and a pinion gear extension that extends within the upper pinion gear orifice wherein the pinion gear extension defines a lower pinion gear extension orifice. The pinion gear extension further defines a side opening. The side opening and the upper pinion gear orifice together define an interference opening that, when occupied, prevents rotation of the upper pinion gear and the lower pinion gear relative to each other. This embodiment also includes a ball bearing in at least one of each side opening, wherein the ball bearing has a diameter greater than a thickness of a wall defining the side opening. A center pin is movable along the pinion gear axis and within the lower pinion gear orifice. The center pin within the lower pinion gear orifice that includes a frustoconical portion between a base portion having a first diameter and a second diameter that is less than the first diameter, and located in the upper pinion gear orifice, whereby movement of the frustoconical portion of the center pin causes radially outward displacement of the ball bearing into the interference opening, thereby causing an interfering relation between rotation of the upper pinion gear relative to the lower pinion gear. A spring at the lower pinion gear provides bias to the center pin radially outward from the longitudinal axis of handle body, whereby the ball bearing is directed radially outward through the side opening into the interference opening, thereby causing the interfering relation of rotation of the upper pinion gear relative to the lower pinion gear, whereby depressing the center pin removes outward displacement of the ball bearing and eliminates the interfering relation between rotation of the upper pinion gear and a lower pinion gear to cause rotation of the proximal handle to be independent of longitudinal movement of the delivery catheter relative to the handle body along the longitudinal axis.

(7) In another embodiment, the delivery device of the invention includes a handle body having longitudinal axis, a proximal end, and a distal end. A gear rack extends within the handle body, and a proximal handle extends about the gear rack and defines teeth, wherein the proximal handle is rotatable about the handle body and the gear rack. A distal handle extends around handle body at the distal end of the handle body. A guidewire catheter having a proximal end and a distal end extends through the handle body, the proximal handle, the distal handle, and along the longitudinal axis. A delivery catheter is axially fixed to the proximal handle and has a distal end extending from within the distal end of the handle body and about the guide wire catheter. An outer catheter extends distally from the distal handle and about the delivery catheter in a first, retracted position. A gear assembly links teeth of the proximal handle to the gear rack, whereby rotation of the proximal handle about the longitudinal axis moves the proximal handle and the delivery catheter along the longitudinal axis relative to the gear rack. The gear assembly includes a pinion gear assembly engaging the gear rack, and a linking gear assembly that includes an upper linking gear engaging the teeth of the proximal handle and a lower linking gear fixed to the upper linking gear and between the upper linking gear and the longitudinal axis of the handle body, and having teeth engaging the pinion gear assembly. The upper linking gear and the lower linking gear have a common axis of rotation that is normal to the longitudinal axis of the handle body, wherein the upper linking gear in the lower linking gear each define a central opening along the common axis of rotation. A push rod extends about the guidewire catheter and within the delivery catheter, and is fixed to the guidewire catheter at the proximal end and to the guidewire catheter proximal to the handle body, and is selectively fixed to the proximal handle. A locking mechanism assembly extends about the push rod and includes a first locking mechanism that locks the delivery catheter to the push rod when the locking mechanism is in a first locking position, wherein the first locking mechanism defines a first socket and the housing defines a second socket, and wherein the opposite

ends of the pin are seated in the first socket and the second socket. A second locking mechanism is fixed to the proximal end of the handle body that locks the push rod to the handle body when the locking mechanism assembly is in a second locking position, wherein the locking function of the first locking position in the second locking position are mutually exclusive. An actuator includes the gear assembly and further includes a housing extending about the handle body, the housing having a proximal end defining a proximal opening and a distal end defining a distal opening, the housing further defining an aperture between the proximal opening and the distal opening. A center pin extends through the central opening of the upper linking gear and the lower linking gear. The pin includes opposite ends that are at the locking mechanism assembly and housing. A clutch engages the gear assembly with the proximal handle and thereby biases longitudinal movement of the gear assembly and associated rotation of the proximal handle about the longitudinal axis, wherein rotation of the proximal handle about the longitudinal axis when the clutch is engaged is resisted by friction between the clutch and at least one of the gear assembly and a portion of the remainder of the delivery device, wherein the clutch engages the pin with the linking gear assembly when the gear assembly is directed in a proximal direction, and wherein rotation of the proximal handle that directs the gear assembly in a proximal direction is resisted by an interfering relationship between the pin and at least one of the first and the second sockets.

(8) In yet another embodiment, the delivery device of the invention includes a handle body having a longitudinal axis, a proximal end, and a distal end, and a gear rack extending within the handle body. A proximal handle extends about the gear rack and defines teeth. The proximal handle is rotatable about the handle body and the gear rack. A distal handle extends around the handle body at the distal end of the handle body and a guidewire catheter has a proximal end and a distal end. The guide wire extends through the handle body, the proximal handle, the distal handle, and along the longitudinal axis. A delivery catheter is axially fixed to the proximal handle and has a distal end extending from within the distal end of the handle body and about the guidewire catheter. An outer catheter extends distally from the distal handle and about the delivery catheter in a first, retracted position. A gear assembly links teeth of the proximal handle to the gear rack, whereby rotation of the proximal handle about the longitudinal axis moves the proximal handle and a delivery catheter along the longitudinal axis relative to the gear rack. The gear assembly includes a pinion gear assembly engaging the gear rack, and a linking gear assembly including an upper linking gear engaging the teeth of the proximal handle and a lower linking gear fixed to the upper linking gear and between the upper linking gear and the longitudinal axis of the handle body, and has teeth engaging the pinion gear assembly. The upper linking gear and the lower linking gear have a common axis of rotation that is normal to the longitudinal axis of the handle body, wherein the upper linking gear in the lower linking gear each define a central opening along the common axis of rotation. A push rod extends about the guidewire catheter and within the delivery catheter, and is fixed to the guidewire catheter at the proximal end of the guidewire catheter proximal to the handle body. The push rod is selectively fixed to the proximal handle. A locking mechanism assembly extends about the push rod and includes a first locking mechanism that locks the delivery catheter to the push rod when the locking mechanism is in a first locking position, and a second locking mechanism fixed to the proximal end of the handle body that locks the push rod to the handle body when the locking mechanism assembly is in a second locking position, wherein the locking function of the first locking position and the second locking position are mutually exclusive. An actuator includes the gear assembly and further includes a housing extending about handle body and has a proximal end defining a proximal opening and a distal end defining a distal opening. The housing further defines an aperture between the proximal opening and the distal opening. A center pin of the actuator extends through the central openings of the upper linking gear and the lower linking gear and includes opposite ends that are at the locking mechanism and the housing. A clutch engages the gear assembly with the proximal handle and thereby biases longitudinal movement of the gear assembly and associated rotation of the proximal handle about the longitudinal axis,

wherein rotation of the proximal handle about the longitudinal axis when the clutch is engaged is resisted by friction between the clutch and at least one of the gear assembly and a portion of the remainder of the delivery device, wherein the clutch selectively engages the pin with the linking gear assembly when the gear assembly is directed in a proximal direction. A second clutch engages the linking gear assembly with the pin when the pinion gear assembly is directed distally along the gear rack during engagement of the linking gear assembly with the pinion gear assembly. The second clutch is a coil spring that is fixed at one end to the first locking mechanism and the pin extends through the coil spring, wherein the coil spring is engaged with the pin when the pinion gear assembly is directed distally along the gear rack during engagement of the linking gear assembly with the pinion gear assembly.

(9) In still another embodiment of the invention, a delivery device includes a handle body having a longitudinal axis, a proximal end, and a distal end. A gear rack extends within the handle body and a proximal handle extends about the gear rack and defines teeth. The proximal handle is rotatable about the handle body and the gear rack. A distal handle extends around handle body at the distal end of the handle body. A guidewire catheter has a proximal end and a distal end, and extends through the handle body, the proximal handle, the distal handle, and along the longitudinal axis. A delivery catheter is axially fixed to the proximal handle and has a distal end extending from within the distal end of the handle body and about the guidewire catheter. An outer catheter extends distally from the distal handle and about the delivery catheter in a first, retracted position. A gear assembly links the teeth of the proximal handle to the gear rack, whereby rotation of the proximal handle about the longitudinal axis moves the proximal handle and the delivery catheter along the longitudinal axis relative to the gear rack. The gear assembly includes a pinion gear assembly engaging the gear rack, and a linking gear assembly that includes an upper linking gear engaging the teeth of the proximal handle, and a lower linking gear that is fixed to the upper linking gear and between the upper linking gear and the longitudinal axis of the handle body. The lower linking gear includes teeth engaging the pinion gear assembly. The upper linking gear and the lower linking gear have a common axis of rotation that is normal to the longitudinal axis of the handle body, wherein the upper linking gear and the lower linking gear each define a central opening along the common axis of rotation. A push rod extends about the guidewire catheter and within the delivery catheter, and is fixed into the guidewire catheter at the proximal end of the guidewire catheter proximal to the handle body, and is selectively fixed to the proximal handle. A locking mechanism extends about the push rod and includes a first locking mechanism that locks the delivery catheter to the push rod when the locking mechanism is in a first locking position, and a second locking mechanism fixed to the proximal end of the handle body that locks the push rod to the handle body when the locking mechanism assembly is in a second locking position, wherein the locking function of the first locking position and the second locking position are mutually exclusive. An actuator that includes the gear assembly further includes a housing extending about the handle body. The housing having a proximal end defining a proximal opening and a distal end defining a distal opening, the housing further defining an aperture between the proximal opening in the distal opening. A center pin of the actuator extends through the central openings of the upper linking gear and the lower linking gear, the pin including opposite ends that are at the locking mechanism and the housing. A clutch at the gear assembly engages the gear assembly with the proximal handle and thereby biases longitudinal movement of the gear assembly and associated rotation of the proximal handle about the longitudinal axis, wherein rotation of the proximal handle about the longitudinal axis when the clutch is engaged is resisted by friction between the clutch and at least one of the gear assembly and a portion of the remainder of the delivery device, and wherein the clutch selectively engages the pin with the linking gear assembly when the gear assembly is directed in a proximal direction. A second clutch that engages in linking gear assembly with the pin when the pinion gear assembly is directed distally along the gear rack during engagement of the linking gear assembly with the pinion gear assembly, wherein the second clutch is a coil spring that is fixed at

one end into the housing and the pin extends through the coil spring, wherein the coil spring is engaged with the pin when the pinion gear assembly is directed distally along the gear rack during engagement of the linking gear assembly with the pinion gear assembly.

(10) The delivery device and method of its use of the invention have many advantages. For example, longitudinal expansion of a prosthesis after removal of force that advances a prosthesis from a sheath, such as can occur upon release of a handle rotated about a delivery device to control delivery to a surgical site, is minimized or eliminated by a clutch. The clutch engages only when a gear assembly that translates rotational force of a handle and provides mechanical advantage for a surgeon to advance to the prosthesis along a longitudinal axis of the delivery device. The resistance to proximal movement, or longitudinal expansion of the prosthesis, can be overcome, again by mechanical advantage, applied by rotation of the proximal handle by the surgeon in the opposite direction to that of advancement. The resistance can be overcome by, for example, static friction between the clutch and at least one of a pin about which the clutch extends and an interference fit between the clutch or the pin and another component of the delivery device that does not rotate with rotation of the handle or longitudinal advancement of the prosthesis to the delivery site. Selective engagement of the clutch and static friction can thereby provide greater control to the surgeon during advancement of the prosthesis to a surgical site and while targeting landing of the prosthesis at the surgical site prior to deployment and release from the delivery device. The clutch and selective employment of static friction, either separately or in combination, can also reduce the likelihood of interference and jamming of the gears that would otherwise cause the delivery device to seize and interrupt or prevent delivery of the prosthesis altogether.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of one embodiment of the delivery device of the invention.
- (2) FIG. 2 is a perspective view of one embodiment of a shifting knob, driveshaft and actuator, of the invention.
- (3) FIG. 3 is a perspective view of the shifting knob and driveshaft of the embodiment shown in FIG. 2.
- (4) FIG. 4 is a perspective view of a shifting knob, distal handle, distal handle nose, and a cross-sectional view, in part, of a handle body and delivery catheter of another embodiment of the invention.
- (5) FIG. 5 is a partial cutaway section of the portion of the embodiment of the delivery device of the invention shown in FIG. 4.
- (6) FIG. 6 is a partial cutaway section of a detail of a portion of the handle body, intermediate gear, reduction gear and connecting gear, all of which link the shifting knob with the driveshaft of the embodiment of the invention shown in FIG. 4.
- (7) FIG. 7 is a partial cutaway view of the embodiment of FIG. 4, showing a cross-sectional view of the distal handle and a base to which outer catheter is connected at distal handle nose.
- (8) FIG. 8 is a partial cutaway, of the embodiment of FIG. 4 showing constricting rings extending about a delivery catheter.
- (9) FIG. 9 is another embodiment of a partial cutaway of the delivery device of FIG. 1 showing an actuator and a push button at the proximal end of a slot defined by the handle body.
- (10) FIG. 10 is a perspective view of first locking component, and second locking component, and their relation to the drive shaft of the embodiment shown in FIG. 1.
- (11) FIG. 11 is another representation of a first locking component and a second locking component, and a first locking component housing and a second locking component housing stabilizing the spatial relation between the first locking component and the second locking

component, respectively, relative to the drive shaft of the embodiment of FIG. 1.

(12) FIG. 12A is another perspective view of the embodiment of FIG. 1, showing displacement of the proximal handle and the actuator along the handle body consequent to rotating of the proximal handle about the handle body or depressing the push button of the actuator to thereby allow longitudinal movement of the actuator and the proximal handle without rotation of the proximal handle.

(13) FIG. 12B is another perspective view of the embodiment of FIG. 1, wherein a proximal handle has been advanced along the handle body of the delivery system.

(14) FIG. 13 is a detail of the proximal handle and the actuator at the handle body of the embodiment of the invention shown in FIG. 1, without the actuator housing.

(15) FIG. 14 is a perspective view of the detail of FIG. 13, without the push button of the actuator shown in FIG. 13.

(16) FIG. 14A is an enlarged view, in perspective, of a linking gear assembly associated with the actuator of FIG. 14.

(17) FIG. 15 is a partial cutaway of the embodiment of FIG. 1 showing the relation of the pinion and the linking gear assemblies relative to the first locking component housing and the relationship of the first locking component housing to the delivery catheter within the housing.

(18) FIG. 16 is an exploded view, in perspective, of a first locking mechanism, a linking gear, a pin, and a coil spring clutch of one embodiment of a delivery device of the invention.

(19) FIG. 17 is a partial cut-away view, in perspective, of a housing, and the first locking mechanism, linking gear, pin, and coil spring of FIG. 16, when assembled.

(20) FIG. 18 is an exploded view, in perspective, of a first locking mechanism, a one-way needle roller bearing clutch and pin of another embodiment of a delivery device of the invention.

(21) FIG. 19 is a side view of a linking gear assembly, a pin and a one-way needle roller bearing clutch at a base of a pin, as assembled in another embodiment of the delivery device of the invention.

(22) FIG. 20 is a side view of a linking gear assembly, a pin and a one-way needle roller bearing clutch at a top portion of a pin, as assembled in still another embodiment of the delivery device of the invention.

(23) FIG. 21 is an exploded view, in perspective, of a linking gear, a pin, a one-way needle roller bearing clutch as a first clutch, and a coil spring as a second clutch, in still another embodiment of the delivery device of the invention.

(24) FIG. 22 is a perspective view of the components shown in FIG. 21 in assembled form, wherein the pin is aligned with a socket of a first locking mechanism of a delivery device of the invention.

(25) FIG. 23 is a side view of the assembled linking gear, pin, clutches, and aligned first locking mechanism of FIG. 22.

(26) FIG. 24 is a plan view, of the interior portion of a housing component of one embodiment of the delivery device of the invention, indicating the location where an end of the spring coil second clutch would be seated when assembled.

(27) FIG. 25 is an exploded view of a linking gear assembly of the invention that includes a first clutch that is a one-way needle roller bearing clutch and a second clutch that is a coil spring.

(28) FIG. 26 is a perspective view, exploded in part, of the linking gear assembly of FIG. 25 in combination with a first locking component housing of the invention.

(29) FIG. 27 is a side view of the combination of the linking gear assembly and first locking component housing of FIG. 26.

(30) FIG. 28 is a plan view of the first locking component housing of FIG. 27.

(31) FIG. 29 is an exploded view of another embodiment of a linking gear assembly of the invention including two one-way needle roller bearing clutches.

(32) FIG. 30 is a side view of the linking gear assembly of the invention shown in FIG. 29 when assembled.

(33) FIG. 31 is a perspective view of the first locking component housing and the second locking component housing within a cutaway view of the handle body, along with a perspective view of the linking gear assembly and the pinion gear assembly of the actuator.

(34) FIG. 32 is a side view of the representation of the invention, as shown in FIG. 16.

(35) FIG. 33 is a partial cutaway of the distal end of handle body and second locking component shown in FIGS. 31 and 32.

(36) FIG. 34 is a perspective view of a partial cutaway of the actuator shown in FIG. 32.

(37) FIG. 35 is a perspective view of a rack and proximal handle of the embodiment shown in FIG. 1, and an alternate embodiment of the actuator of the invention, lacking a linking gear assembly.

(38) FIG. 36 is a perspective view, partially transparent, of the embodiment of the pinion gear assembly of FIG. 35.

(39) FIG. 37 is another view of the embodiment represented in FIG. 36.

(40) FIG. 38 is a perspective view of the embodiment shown in FIGS. 36 and 37, lacking the upper pinion gear shown in those figures.

(41) FIG. 39 is another embodiment of the representation shown in FIG. 38.

(42) FIG. 40A is a cross-sectional view of the pinion gear assembly of FIGS. 36 through 39, wherein a pin has not been depressed, thereby causing the upper pinion gear and lower pinion gear to be in interfering relation to each other by radially outward displacement of ball bearings, and further showing a lower extension of a lower pinion gear of the pinion gear assembly, and a one-way needle roller clutch in which the lower extension is seated.

(43) FIG. 40B is a cross-sectional view of the pinion gear assembly of FIG. 40A, wherein a center pin has been depressed, thereby compressing a biasing spring and allowing independent rotation of upper and lower pinion gears by removal of radially outward displacement of ball bearings that otherwise would cause the interfering relationship of independent rotation of the upper and lower pinion gears.

(44) FIG. 40C is a cross-sectional view of the pinion gear assembly of FIGS. 36 through 39, wherein a pin has not been depressed, thereby causing the upper pinion gear and lower pinion gear to be in interfering relation to each other by radially outward displacement of ball bearings, and further showing an upper extension of a lower pinion gear of the pinion gear assembly, and a one-way needle roller clutch in which the upper extension is seated.

(45) FIG. 40D is a cross-sectional view of the pinion gear assembly of FIG. 40C, wherein a center pin has been depressed, thereby compressing a biasing spring and allowing independent rotation of upper and lower pinion gears by removal of radially outward displacement of ball bearings that otherwise would cause the interfering relationship of independent rotation of the upper and lower pinion gears.

(46) FIG. 41 is an exploded view, in perspective of the pinion gear assembly of FIG. 40, in alignment with a first locking mechanism defining a socket into which the one-way needle roller bearing clutch is seated, such as by a press-fit of the one-way needle roller bearing clutch into the socket.

(47) FIG. 42 is a perspective view of the linking gear assembly of FIGS. 40 and 41 in perspective.

(48) FIG. 43 is a side view of the assembled linking gear assembly of FIG. 40 through 42 when assembled with the first locking mechanism of FIG. 41.

(49) FIG. 44 is a perspective view of one embodiment of a proximal clasp assembly of one embodiment of the invention.

(50) FIG. 45 is a partial cutaway of the proximal clasp assembly shown in FIG. 44.

(51) FIGS. 46A-46C are perspective sectional views of the distal end of the delivery device shown in FIG. 1.

(52) FIG. 47A is a perspective view of the shifting knob in the first position, wherein the push rod is fixed to the proximal handle and the prosthesis is undeployed.

(53) FIG. 47B is a detailed perspective view of proximal clasp assembly in a first position,

whereby the apex clasp assembly is unopened.

(54) FIG. 47C is a detailed perspective view of the shifting knob in the first position.

(55) FIG. 48A is a perspective view of the delivery device of FIGS. 47A through 47C showing advancement of the delivery sheath containing the prosthesis when the shifting knob is in a second position, wherein the push rod is fixed to the handle body.

(56) FIG. 48B is a detailed perspective view of advancement of the delivery sheath of FIG. 48A.

(57) FIG. 49A is a perspective view of the delivery device of FIGS. 48A, 48B showing advancement of the delivery sheath.

(58) FIG. 49B is a detailed perspective view of the shifting knob of FIG. 49A in a second position.

(59) FIG. 50A is a perspective view of the delivery device of FIGS. 49A, 49B, wherein the delivery sheath has been partially retracted from the prosthesis.

(60) FIG. 50B is a representation of an apex clasp assembly of one embodiment of the invention in a closed position.

(61) FIG. 51A is a perspective view of the delivery device of FIG. 50A, wherein the apex clasp assembly is opened by actuation of the proximal clasp assembly to thereby release the apices of the proximal stent of the prosthesis shown in FIG. 51C.

(62) FIG. 51B is a representation of the proximal clasp assembly of FIGS. 44, 45, whereby an apex clasp assembly, not shown, has been opened.

(63) FIG. 51C is a representation of the apex clasp assembly of one embodiment of the invention, in an open position.

(64) FIG. 52A is a perspective view of the delivery device of FIG. 51A, wherein the shifting knob has been moved to the third position, whereby the push rod has been released from the proximal handle and the handle body and, wherein the push rod has been retracted from the fully deployed prosthesis.

(65) FIG. 52B is a perspective view of the shifting knob in the third position as shown in FIG. 52A.

DETAILED DESCRIPTION OF THE INVENTION

(66) While this invention has been particularly shown and described with references to example embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the invention encompassed by the appended claims.

(67) One embodiment, of the delivery device **10** of the invention is shown in FIG. **1**. Delivery device **10** includes guidewire catheter **12** (FIGS. **10**, **11**) having a proximal end **14** and a distal end **16**. "Proximal," as a term employed herein with reference to the delivery device and its components, means relatively close to the surgeon operating the delivery device. "Distal," as a term employed herein with reference to the delivery device and its components, means relatively distal from the surgeon operating the delivery device. "Proximal," as a term employed herein with reference to the prosthesis, stent-graft and components, means relatively close to the heart of the patient. "Distal," as a term employed herein with reference to the prosthesis, stent-graft and components, means relatively distal from the heart of the patient. Returning to FIG. **1**, delivery device **10** includes delivery assembly **18** that extends about the guidewire catheter (not shown). Delivery assembly **18** includes handle body **20** having major longitudinal axis **22**, proximal end **24** and distal end **26**. Delivery catheter **28** (FIG. **9**) has distal end **30** (FIG. **27B**) extending from within distal end **26** of handle body **20** (FIG. **1**) and about the guidewire catheter (not shown).

(68) In one embodiment, push rod **32** extends about guidewire catheter **12** and within delivery catheter **28** (FIGS. **10**, **11**). Push rod **32** is fixed to guidewire catheter **12** at proximal end **34** of push rod **32** proximal to the handle body at pin **192** (FIG. **25**). Referring back to FIG. **1**, proximal handle **36** extends about handle body **20** and is axially fixed to delivery catheter **28**. Proximal handle **36** is selectively fixed to push rod **32**, wherein proximal handle **36** is rotatable about handle body **20** and rotation of proximal handle **36** about handle body **20** translates to longitudinal movement of delivery catheter **28** along longitudinal axis **22** and, selectively, of push rod **32** relative to handle

body **20**, as can be seen by comparing FIG. **12A** with FIG. **12B**. First locking mechanism **38** (FIG. **15**) at handle body **20** selectively engages proximal handle **36** (FIGS. **12A** and **12B**) with push rod **32**.

(69) Distal handle **40** extends about handle body **20** at distal end **26** of handle body **20** and is distal to shifting knob **42** of locking mechanism **37**, that includes first locking mechanism **38** and second locking mechanism **132** (FIG. **15**). Distal handle nose **44** (FIG. **1**) extends distally from distal handle **40** and includes flush port **46** for providing fluid communication between a solution source (not shown) and interior components of delivery device **10**, as necessary, to hydrate contact between components of delivery device **10** and a vascular prosthesis (not shown) within a subject before implantation of the vascular prosthesis in a subject. Outer catheter **48** extends from distal handle nose **44** (FIG. **1**).

(70) Actuator **80** is linked to proximal handle **36**, whereby proximal handle **36** can rotate about handle body **20** while push-button **82** at housing **81** of actuator **80** remains aligned with slot **84** defined by handle body **20**. Depression of push-button **82** of actuator **80** selectively disengages rotation of proximal handle **36** from handle body **20**, whereby rotation of proximal handle **20** is independent of longitudinal movement of delivery catheter **28** relative to handle body **20** along longitudinal axis **22**.

(71) As can be seen in FIG. **2**, shifting knob **42** is linked to drive gear **86** by drive shaft **88**. Drive shaft **88** has proximal end **90** and distal end **92**, and runs along the interior of the handle body **20** (not shown). As can be seen in FIG. **3**, shifting knob **42** is linked to drive shaft **88**, in one embodiment, by intermediate gear **94A**, whereby rotation of shifting knob **42** about handle body **20** causes rotation of drive shaft **88** by virtue of linkage between shifting knob **42** and drive shaft **88** by intermediate gear **94A**. In this embodiment, shifting knob **42** is linked to drive shaft **88** indirectly, as opposed to direct linkage. "Direct linkage" is an optional embodiment, which would be direct contact between shifting knob **42** and drive shaft **88**. Shifting knob **42** is rotatably linked to distal handle **40**, which is fixed to distal end **26** of handle body **20**, as shown in FIG. **1**.

(72) In another embodiment, shown in FIGS. **4** and **5**, linkage between shifting knob **42** and drive shaft **88** includes a gear reduction at intermediate gear **94B** that is linked to coaxial reduction gear **96** which, in turn, is linked to connecting gear **98** that is coaxially linked to drive shaft **88**. By virtue of the gear reduction, the rate of rotation of shifting knob **42** relative to drive shaft **88** can be controlled by the relative dimensions of reduction gear **96** and connecting gear **98** (FIGS. **5**, **6**, **7**). Typically, the rotation ratio, or reduction ratio, of shifting knob **42**:drive shaft **88** is in a ratio of between about 1:2 and about 1:6. The relationship between reduction gear **96** and connecting gear **98** can be seen in greater detail in FIG. **6**.

(73) As can be seen in greater detail in FIG. **7**, delivery catheter **28** extends through handle body **20**, distal handle **40** and distal handle nose **44**. Referring back to FIG. **5**, outer catheter **48** is linked to base **102**, whereby outer catheter **48** is rotatable independently of handle body **20**. As shown in FIG. **8**, constricting rings **104** extend along the delivery catheter **28** within handle body **20**. As shown in FIGS. **8** and **9**, constricting rings **104** have an outside diameter greater than the width of slot **84**, whereby constricting rings **104** will prevent application of longitudinal compressive force by proximal handle **36** on delivery catheter **28** from causing delivery catheter **28** to buckle and thereby move through slot **84** and outside of handle body **20**. Constricting rings **104** also have an inside diameter slightly less than the outside diameter of delivery catheter **28**, whereby constricting rings **104** will have an interference fit with delivery catheter **28**, so that constricting rings **104** can move longitudinally along delivery catheter **28** if directed, but otherwise will remain in place relative to delivery catheter **28**. Gear rack **106** extends longitudinally within handle body **20**. Pin **108** at distal end **26** of handle body **20** extends from distal end **26** of handle body **20** and is selectively slotted within slots **110,112,114** of shifting knob **42**. Shifting knob **42** is longitudinally moveable along handle body **20** and is rotatable about handle body **20** sufficient to allow rotation of shifting knob **42** to move placement of pin **108** within any of slots **110,112,114** of shifting knob

42, which thereby causes rotation of intermediate gear **94**. As a consequence, drive shaft **88** rotates about longitudinal axis **116** (FIG. **10**) of drive shaft **88**. Shifting knob **42** is biased against pin **108** by spring **118** (FIG. **7**).

(74) As can be seen in FIG. **9**, gear rack **106** and drive shaft **88** extend the length of slot **84**. FIG. **10** shows the relation between drive shaft **88**, push rod **32** and first locking mechanism **38**. Push rod **32** extends through first locking mechanism **38** which, in turn, is engaged with drive shaft **88** at drive gear **86** of first locking mechanism **38**. First locking mechanism **38** is fixed relative to proximal handle (not shown) at distal bearings **120** through which push rod **32** extends. Distal bearings **120** are linked to first locking component housing **150** by pins **122**. First locking component **124** of first locking mechanism **38** is fixed relative to distal bearings **120** at distal end **126** and linked to drive gear **86** at proximal end **128**, whereby rotation of drive shaft **88** and consequent rotation of drive gear **86** will further coil, or reduce coil, of first locking component **124**, resulting in engagement or disengagement, respectively, of locking mechanism **38** and, consequently, proximal handle (not shown), with push rod **32**. When first locking mechanism **38** is engaged with push rod **32**, longitudinal movement of proximal handle (not shown) along drive shaft **88** and, thus, handle body **20**, will cause longitudinal movement of push rod **32** along drive shaft **88** and handle body **20**, as can be seen by comparing FIGS. **12A** and **12B**.

(75) Referring back to FIGS. **10**, **11**, drive shaft **88** is rotatably fixed to handle body **20** (FIG. **9**) at driveshaft bearing **130**, which is part of second proximal locking component housing **152** at proximal end **90** of drive shaft **88**. Second locking mechanism **132** includes translating gear **134** that is engaged with drive shaft **88** at proximal end **90** of the drive shaft **88** and is rotatably engaged with mechanism bearings **136** (FIG. **11**), including proximal bearing **138** (FIG. **10**) and distal bearing **140** (FIG. **10**) which, in turn, are fixed relative to handle body **20** at pins **142**. Proximal bearing **138** is radially and axially fixed to handle body **20**. Distal bearing **140** is axially fixed to handle body **20**. Second locking component **144** of second locking mechanism **132** is engaged with one of proximal bearing **138** at proximal end **146** of second locking component **144**, and engaged with translating gear **134** at distal end **148** of second locking component **144**, whereby rotation of drive shaft **88** and, consequently, rotation of translating gear **134** will tighten and engage, or loosen and disengage, second locking component **144** with push rod **32**. When engaged with push rod **32**, second locking component **144** causes push rod **32** to be fixed in location relative to handle body (not shown). When loosened and disengaged from push rod **32**, push rod **32** is longitudinally movable relative to handle body (not shown). The orientation of first locking component **124** and second locking component **144** are reversed, whereby rotation of drive shaft **88** in one direction will, simultaneously, cause engagement and disengagement of first locking component **124** and second locking component **144** with push rod **32**, respectively. Disengagement of first locking component **124** from push rod **32** is caused by movement of shifting knob **42** from a first position defined by pin **108** at slot **110** of shifting knob **42** to second position **112**, defined by pin **108** at second slot **112** of shifting knob **42** (FIG. **9**). The same movement from the first to second position of shifting knob **42** will simultaneously cause engagement of second locking component **144** with push rod **32**, whereby push rod **32** will be fixed in position relative to handle body **20** at second locking component **144** regardless of movement of proximal handle **36** along longitudinal axis **116** of handle body **20**. Referring back to FIGS. **8** and **9**, positioning shifting knob **42**, so that pin **108** is at intermediate slot **114** between the first slot **110** and second slot **112** of shifting knob **42**, will cause both first locking component **124** and second locking component **144** to be disengaged from push rod **32**.

(76) As can be seen in FIG. **11**, first locking component housing **150** fixes lateral movement of first locking component **124** and drive shaft **88**, and second locking component housing **152** fixes the position of second locking component **144** and bearings **138,140** relative to proximal end **90** of drive shaft **88**, respectively. Further, as can also be seen in FIG. **11**, apex release catheter **154** extends within push rod **32** and guidewire catheter **12** extends within apex release catheter **154**.

(77) FIGS. 12A and 12B indicate relative movement of actuator 80 and proximal handle 36 along handle body 20. Rotation of proximal handle 36 about handle body 20, when push button 82 is in a first position, as shown in FIGS. 12A and 12B, will cause longitudinal movement of proximal handle 20 and actuator 80 along handle body 20. Upon depression of push button 82 to a second position essentially flush with actuator housing 81, rotation of proximal handle 36 will not cause longitudinal movement of proximal handle 36 or actuator along handle body 20. Rather, proximal handle 36 and actuator 80 will be movable along handle body 20 without rotation of proximal handle 36 about handle body 20.

(78) As can be seen in FIGS. 13-15, teeth 156 of proximal handle 36 engage upper linking gear 160 of linking gear assembly 158. Linking gear assembly 158 is engaged with pinion gear assembly 164. Lower linking gear 162 of linking gear assembly 158 engages upper pinion gear 166 of pinion gear assembly 164. Pinion gear assembly 164 is linked to first locking component housing 150 (FIG. 11) through slot 84. Linking gear assembly 158 and pinion gear assembly 164 are components of actuator 80, referenced with respect to FIG. 1. As can be seen in FIGS. 14 and 14A, coil spring 210 is seated about pin 214 and is fixed at end 212 within recess 216 of upper linking gear 160.

(79) FIG. 16 is an exploded view of first locking component housing 150, pin 214, upper linking gear 160 and coil spring 210, in perspective, showing how coil spring 210 is wrapped about, or around, pin 214, and extension 218 at one end 212 of coil spring 210 is hooked to allow coil spring 210 to be fixed at opening 217 of upper linking gear 160 against rotation about pin 214 when upper linking gear 160 is rotated in a direction that causes spring 210 to tighten about pin 214. In this case, pin 214 is aligned with, and to be seated in socket 174 (FIG. 11) of first locking component housing 150 when assembled. Pin 214 is in sufficient contact with coil spring 210 when assembled to cause rotation of upper linking gear 160 in one direction to tighten coil spring 210 about pin 214, thereby causing the coil spring 210 to act as a clutch, where further rotation of the linking gear assembly 158 will also cause rotation of pin 214. For example, as shown in FIG. 16, rotation of upper linking gear 160 in a clockwise direction about pin 214 will cause coil spring 210 to tighten about pin 214, thereby locking with pin 214 as a clutch mechanism, causing pin 214 to rotate with upper linking gear assembly 158 upon continued clockwise rotation of upper linking gear 160. This clutching mechanism, whereby pin 210 tightens about pin 214 and locks with it, can occur, for example, by direction of first locking mechanism 38 (FIG. 15) in a proximal direction (toward the surgeon) by longitudinal expansion of a vascular prosthesis 58 (FIG. 47A, infra) during implantation prompted by release of proximal handle 36 by the surgeon after having advanced the prosthesis to a surgical site. Conversely, when proximal handle 36 is rotated by the surgeon in a clockwise direction, upper linking gear 160, the teeth of which are engaged with teeth of proximal handle 36, will rotate in counterclockwise direction, thereby causing coil spring 210 to expand in diameter about pin 214 and thereby release pin 214 from rotation of upper linking gear 160.

(80) FIG. 17 is a partial cut-away view, in perspective, of a portion of housing 81 and first locking mechanism 38, in combination with linking gear assembly 158. Pinion gear assembly 164 is not shown for the purpose of clearly showing linking gear assembly 158. As can be seen from the embodiment of the invention shown in FIG. 17, pin 214 extends through linking gear assembly 158 and is secured at one end 220 in socket 222 defined by housing 81 and at opposite end 224 in socket 174 defined by first locking mechanism 38. In this embodiment, pin 214 is seated in at least one of socket 222 and socket 174 with an interference fit, whereby pin 214 will resist rotation about its longitudinal axis. The resistance is sufficient to prevent longitudinal expansion of a stent graft to be delivered along longitudinal axis 22 (FIG. 1) when proximal handle 36 (FIGS. 1, 13, 14) is released by the surgeon after at least partial advancement of a vascular prosthesis, thereby preventing backspin. The resistance to rotation of pin 214 is not so great, however, that it cannot be overridden by the surgeon in the event the vascular prosthesis is to be intentionally moved back toward the surgeon (prior to deployment of the vascular prosthesis) by rotating proximal handle 36

in the opposite direction to that of advancement (such as in a counterclockwise direction, which is opposite to the clockwise direction of rotation of the proximal handle **36** described above). In one embodiment, the amount of force exhibited as resistance to rotation by housing **81** is in a range of between about 2.0 lbf. inch and about 12.0 lbf. inch. In another embodiment, the amount of force exhibited as resistance to rotation by housing **81** is in a range of between about 5.0 lbf. inch and about 7.0 lbf. inch. Preferably, housing **81**, or at least the portion of housing **81** defining socket **222**, is fabricated of medical grade engineering plastic that is gamma radiation-compatible. Housing **81** preferably is an injection molded engineering plastic. Socket **222** may also be injection molded engineering plastic. Preferably, first locking mechanism **38** defining socket **174** is fabricated of stainless steel, anodized aluminum or medical grade engineering plastic. Preferably, only socket **222** of housing **81** provides resistance to rotation of pin **214**. Depending on where resistance to backspin is to be overcome in the delivery device of the invention, at least one of sockets **174**, **222**, pin **214**, or the needles of a one-way needle roller bearing clutch, described below, can be formed of stainless steel in order to control, in part the torque force required to overcome the static friction of the interference fit. In another embodiment, not shown, where a second clutch is employed, such as at socket **174**, the second clutch provides resistance to rotation of pin **214**.

(81) In another embodiment of the invention, shown in FIG. **18**, coil spring **214** of FIGS. **16** and **17** is replaced by one-way needle roller bearing clutch **226**, such as is known in the art. A one-way needle roller bearing clutch is a clutch that includes “needles” aligned on the inside surface of a cylinder, where the needles roll freely about a pin that is rotated within the cylinder in one direction, but lock, and thereby provide torque when the pin is directed in an opposite direction of rotation about its axis. FIG. **18** is an exploded view, in perspective, of first locking mechanism **38**, pin **214**, and linking gear assembly **158**, wherein, instead of coil spring **214**, one-way needle roller bearing clutch **226** is employed. In this embodiment, when assembled, one-way needle roller bearing clutch **226** will be press fit into orifice **240** defined by upper and lower linking gears of linking gear assembly **158**. Like the embodiment shown in FIG. **16**, the embodiment of FIG. **18** allows clockwise rotation of proximal handle **36**, such as is shown in FIGS. **13** and **14**, and consequent counterclockwise rotation of linking gear assembly **158** about pin **214**, which is in an interference fit with at least one of socket **222** defined by housing **81**, and socket **174** of first locking component housing **150** of first locking mechanism **38**, to thereby advance a vascular prosthesis to a surgical site. One-way needle roller bearing clutch **226**, however, will lock about pin **214** and resist, by virtue of the interference fit between pin **214** and at least one of socket **222** and socket **174**, proximal movement of gearing assembly **230** (FIGS. **13** and **14**), consisting of linking gear assembly **158** and pinion gear assembly **164** toward the surgeon, such as would be a consequence a longitudinal expansion of a vascular prosthesis when the surgeon releases proximal handle **36** (FIG. **13** and **14**) after at least partially advancing the vascular prosthesis toward a surgical site. Like the embodiment shown in FIG. **16**, the resistance to rotation of pin **214** provided by the interference between pin **214** and at least one of socket **222** and socket **174**, can be overcome by the surgeon by forcefully rotating proximal handle **36** in a counterclockwise direction to thereby retract the vascular prosthesis back toward or within the delivery device. It should be understood that, in alternative embodiments, the component parts of the delivery device can be constructed so that the functions described above can be performed by counterclockwise rotation where clockwise rotation is employed above and by clockwise rotation where counterclockwise rotation is employed. Also, it is to be understood that other arrangements of resistance to rotation of linking gear assembly **158** when one-way needle roller bearing clutch **226** is locked can be employed. For example, in an embodiment where pin **214** is fused with, or locked in place at either housing or first locking mechanism, an interference fit that would resist backspin, but enable a surgeon to retract a vascular prosthesis by rotating proximal handle **36** in a direction about delivery device in a direction opposite to that of advancement could be achieved by, for example, an

interference fit between one-way needle roller bearing clutch **226** and linking gear assembly **158**, or by overriding friction between pin **214** and component needles within the one-way needle roller bearing clutch **226**.

(82) FIGS. **19** and **20** show a side view and a perspective view, respectively, of other embodiments of a linking gear assembly of the invention in combination with a one-way needle roller bearing clutch. In this embodiment, pin **214** is either fused with, or in an interference fit with upper linking gear **160** and lower linking gear **162**. In FIG. **19**, one-way needle roller bearing clutch **232** is at one end of pin **214** and, while not shown, one-way needle roller bearing clutch **232** will be seated in socket **174** of first locking component housing **150**, such as by being press-fit into socket **174**. In FIG. **20**, one-way needle roller bearing clutch **234** is at the opposite end of pin **214** and, while again not shown, one-way needle roller bearing clutch **234** will be seated in socket **222** of housing, such as by being press-fit into socket **222** of housing **81** (FIG. **17**). In either case, resistance to backspin when first locking component housing **150** is urged in a proximal direction (toward the surgeon) (FIGS. **12A** and **12B**), as a consequence of the surgeon releasing proximal handle **36** after having advanced the vascular prosthesis, can be resisted and caused by any one or a combination of friction between pin **214** and linking gear assembly **158**, pin **214** and one-way needle roller bearing clutch **232** or between one-way needle roller bearing clutch **234** and socket **222** (FIGS. **16** and **17**), and the one-way needle roller bearing clutch **234** and socket, such as socket **174** or **222**, in which it is seated. It is to be understood that the amount of torque applied in resistance to backspin can differ among the components contributing to overall resistance, depending on the application and particular configuration and needs of the delivery device for implantation of a specific vascular prosthesis.

(83) FIGS. **21-30** are views of embodiments of the invention wherein the delivery device of the invention includes two clutches **236**, **238**, each engaging in rotation about pin **214** in an opposite direction to that of engagement of the other about the same pin **214**. In each of these embodiments, pin **214** need not be in an interference fit with the socket of either housing **81** or first locking mechanism **38**. Instead, while first clutch **236**, which can be a one-way roller needle bearing clutch, as shown in FIG. **21**, prevents backspin, as described above, second clutch **238**, such as a coil spring, extending about pin **214** of the linking gear assembly **158**, will engage when the linking gear assembly **158** is engaged with **214** pin and proximal handle **36** (FIGS. **12A** and **12B**) is being rotated to advance the vascular prosthesis in a distal direction away from the surgeon and toward the surgical site. During engagement of second clutch **238** in this embodiment, pin **214** rotates within second clutch **238** while second clutch **238** is engaged, overcoming static friction between second clutch **238** and pin **214**. The torque necessary to overcome resistance to rotation of **214** pin within second clutch **238** while second clutch **238** is engaged can replace that which would result from an interference fit between pin **214** and socket **222** of housing **81** or socket **174** of first locking component housing **150**. Second clutch **238** can thereby improve control by the surgeon during surgery by enabling manufacture of a delivery device that is less dependent upon variable factors in the construction of sockets in either housing **81** or first locking locking component housing **150**, the specifications for which generally require a very low tolerance in order to function within performance limitations during use. Alternatively, pin **214** and linking gear assembly **158** may not engage during rotation of linking gears **160**, **162** that advances the vascular prosthesis. Rather, linking gears **160**, **162** rotate freely about pin **214** when linking gears **160**, **162** rotate in a direction to advance the vascular prosthesis. Such an arrangement can be a consequence, for example, of first clutch **236**, such as one-way needle roller bearing clutch shown in FIGS. **21-24**, or a coil spring between the linking gears and the pin, so that first clutch **236** engages linking gears **160**, **162** with pin **214** only when first locking mechanism **38** is directed proximally (toward the surgeon) by release of proximal handle **36** (FIGS. **12A** and **12B**). In this embodiment, second clutch **238** has no function during advancement of the vascular prosthesis by rotation of proximal handle **36**, but will provide resistance to rotation of pin **214** (FIG. **21**) during any backspin of

linking gears **160**, **162** and proximal handle **36** caused by direction of first locking mechanism **38** in a proximal direction, or by deliberate rotation of proximal handle **36** by the surgeon to retract the vascular prosthesis. In one embodiment, regardless of whether linking gears **160**, **162** rotate freely about pin **214** during advancement of the vascular prosthesis, the force necessary to overcome friction caused by second clutch **238** is less than that caused by first clutch **236**.

(84) In one specific embodiment of the invention that includes two clutches, FIG. **21** is an exploded view of an assembly of the invention that includes a perspective view of pin **214**, upper linking gear **160**, first clutch **236**, which is a one-way needle roller bearing clutch, and second clutch **238**, which is a coil spring clutch. When partially assembled, as shown in FIG. **22**, first clutch **236** is press-fit into orifice **240** defined by upper linking gear **160** and one end **242** of coil spring **238** is locked to housing **81** a slot **244**, shown in FIG. **24**. FIG. **23** is a side view of the partial assembly shown in FIG. **22**. Pin **214** extends through first clutch **236** and second clutch **238**. FIG. **24** is a plan view of housing **81**, indicating slot **244** where end **242** of coil spring **238** will be seated upon assembly of linking gear assembly **158**, pin **214**, coil spring **238** and housing **81**. Upon assembly, opposite ends of pin **214** are seated in socket **222** of housing **81** and socket **174** of first locking component housing **150**, as described above. During advancement of a vascular prosthesis by clockwise rotation of handle **36**, as described above, first clutch **236** will not be engaged, so that upper linking gear **160** and lower linking gear **162** will rotate freely about pin **214**. Upon backspin or clockwise rotation of upper linking gear **160**, such as by urging of linking gear assembly **158** in a proximal direction by longitudinal expansion of a partially longitudinally-compressed vascular prosthesis, first clutch **236** will lock rotation of pin **214** to that of upper linking gear **160**, and rotation of pin **214** within second clutch **238** will loosen clutch about pin **214**, while still providing frictional resistance to rotation of pin **214** within second clutch **238**. In a preferred embodiment, the force necessary to override the frictional resistance to rotation of pin **214** within first clutch **236** when first clutch **238** is locked is greater than that necessary to overcome the frictional resistance to rotation of pin **214** within second clutch **238** when pin **214** is rotated in the same direction that causes the first clutch **236** to lock.

(85) In another specific embodiment of the invention that includes two clutches, FIG. **25** is an exploded view of assembly of the invention that includes a perspective view of pin **214**, linking gear assembly **158**, first clutch **236** that is a one-way needle roller bearing clutch, and second clutch **246** that is a coil spring. When at least partially assembled, first clutch **236** is press-fit into orifice **240** defined by upper linking gear **160**, and coil spring **246** extends about one end of pin **214**, as shown in FIG. **25**. One end **248** of coil spring **246** is locked to first locking component housing **150**, as shown in FIG. **26**, which is a perspective view of a combination of the partial assembly of FIG. **25** in assembled form. FIG. **27** is a side view of the embodiment shown in FIGS. **25** and **26**. FIG. **28**, is a plan view of first locking component housing **150**, showing socket **174** and slot **247** into which end **248** of coil spring **246** fits. Pin **214** extends through first clutch **236** and second clutch **246**, as shown in FIG. **26**. FIG. **28** is a side view of the partial assembly of FIG. **26**. One end of pin **214** and coil spring **246** can be seated, when fully assembled, in socket **174** of first locking component housing **150**. The opposite end of pin **214** is seated in socket **222** of housing **81** (FIG. **17**) as described above. During advancement of a vascular prosthesis by clockwise rotation of handle **36**, as described above, first clutch **236** will not be engaged, so that linking gear assembly **158** will rotate freely about pin **214**. Upon backspin or clockwise rotation of upper linking gear, first clutch **236** will lock rotation of pin **214** to that of upper linking gear **160**, and rotation of pin **214** within second clutch **238** will loosen second clutch (a coil spring, as shown in FIGS. **25-28**) **238** about pin **214**, while still providing frictional resistance to rotation of pin **238**. In a preferred embodiment, the force necessary to override the frictional resistance to rotation of pin **214** within first clutch (which is the one-way needle roller bearing clutch) **236** when first clutch **236** is locked is greater than that necessary to overcome the frictional resistance to rotation of pin **214** within the second clutch **238** when the pin **214** is rotated in the same direction that causes the first clutch **236**

to lock.

(86) In yet another specific embodiment of the invention that includes two clutches, FIG. 29 is an exploded view of another assembly of the invention that includes a perspective view of pin 214, linking gear assembly 158, first clutch 236, that is a one-way needle roller bearing clutch, and second clutch 250, that is a one-way needle roller bearing clutch. In another embodiment, first clutch 236 and second clutch 250 are both disengaged during distal advancement and engaged when directed in a proximal direction. In this alternative embodiment, the force necessary to override the frictional resistance to rotation of pin 214 within first clutch 236 when first clutch 236 is locked can be greater than that necessary to overcome the frictional resistance to rotation of pin 214 within second clutch 250 when pin 214 is rotated in a direction that causes first clutch 236 to lock.

(87) When assembled, as shown in FIG. 30, first clutch 236 is press-fit into orifice 241 of upper linking gear 160 and second one-way needle roller bearing clutch 250 is press-fit into socket 174 of first locking component housing 150 (FIG. 18). It is to be understood that the positions of the first and second one-way needle roller bearing clutches can be reversed. Pin 214 extends through first clutch and second clutch. Opposite ends of pin 214 are seated in socket 222 of housing 81 (FIG. 17) and second one-way needle roller bearing clutch 250. During advancement of a vascular prosthesis by clockwise rotation of handle 36, as described above, first one-way needle roller bearing clutch 236 will not be engaged, but the torque applied to proximal handle by the surgeon will be sufficient to override the frictional resistance provided to pin 214 by second one-way needle roller bearing clutch 250. The frictional resistance can be a consequence of friction between second one-way needle roller bearing clutch 250 and socket 174 of first locking component housing 150, or between the needles of second one-way needle roller bearing clutch 250 and pin 214. Upon backspin or clockwise rotation of upper linking gear 160 (or linking gear assembly 158), first clutch 236 will lock rotation of pin 214 to that of upper linking gear 160, and rotation of rotation of pin 214 within second clutch 250 will unlock second clutch 250 from pin 214. In one embodiment, the surgeon can rotate proximal handle 36 in a direction to retract a vascular prosthesis by overriding the frictional force provided by first one-way needle roller bearing clutch 236. The frictional force can be, for example, at least one of that between the first one-way needle roller bearing clutch 236 and orifice 241 of linking gear assembly 158 into which it is press-fit, and between the needle rollers of the one-way needle roller clutch 236 and pin 214. In a preferred embodiment, the force necessary to override the frictional resistance to rotation of pin 214 within first clutch 236 when first clutch 236 is locked is greater than that necessary to overcome the frictional resistance to rotation of pin 214 within the second clutch 250 when pin 214 is rotated in the direction that causes the first clutch 236 to lock. The relative torque force required to override the friction between pin 214 and the first one-way needle roller clutch 236 and second one-way needle roller clutch 250 can be manipulated, for example, by employing metal needle rollers in first one-way needle roller bearing clutch 236 and plastic needles in second one-way needle roller bearing clutch 250.

(88) FIG. 31 is a perspective view of actuator 80 of FIG. 1 (without housing 81 or pushbutton 82), of first locking component housing 150 and second locking component housing 152. Coil spring 210, as a clutch that extends about pin 214. End 218 of coil spring 210 is seated within recess of upper linking gear. This embodiment of pin 214, clutch 210, linking gear assembly 158 and first locking component housing 150 is the same as that shown in FIG. 16.

(89) In another embodiment, shown in FIG. 32, which employs the assembly shown in FIGS. 25-28, upper pinion gear 166 is coaxial with lower pinion gear 168 which, in turn, engages gear rack 106. As shown therein, first clutch is a one-way needle roller bearing clutch 236 that is press-fit into orifice 240 defined by linking gear assembly 158, and second clutch is a coil spring 246 that is seated in socket 174, and through which pin 214 extends. In both embodiments of FIGS. 31 and 32, delivery catheter 28 is linked to first locking component housing 150 and, thus, will move

longitudinally along housing **150**, with movement of proximal handle **36** and actuator **80**, as shown in FIG. **1**, regardless of whether first locking component **124** (FIG. **19**) is engaged with push rod **32** (FIGS. **33** and **34**). Therefore, when upper pinion gear **166** engages lower pinion gear **168**, rotation of proximal handle **36** (as shown in FIG. **1**) about handle body **20** will cause rotation of linking gear assembly **158** (FIGS. **1** and **34**) and, consequently, rotation of pinion gear assembly **164** (FIG. **34**) and movement of pinion gear assembly **164** along gear rack **106** (FIGS. **31**, **32**, and **34**), and movement of proximal handle **36** (FIG. **1**) and actuator **80** (FIGS. **12A** and **12B**) along handle body **20**. Further, while first locking component **124** (FIGS. **10**, **11**, **33**, and **34**) is engaged with push rod **32**, rotation of proximal handle **36** will cause longitudinal movement of push rod **32** along handle body **20**. In all cases, movement of proximal handle **36** and actuator **80** along handle body **20** will always occur together, and will cause movement of delivery catheter **28** longitudinally along handle body **20**.

(90) However, as will be further explained below, depression of center pin **170** disengages upper pinion gear **166** from lower pinion gear **168**. When upper pinion gear **166** is disengaged from lower pinion gear **168**, rotation of proximal handle **36** about handle body **20** does not cause longitudinal movement of the proximal handle **36** and actuator **80** along handle body **20**. Further, longitudinal movement of proximal handle **36** and actuator **80** along handle body **20** can be obtained simply by moving proximal handle **36** and actuator **80** along handle body **20** without rotation of proximal handle **36** about handle body **20** (FIGS. **1**, **12A** and **12B**).

(91) FIG. **34** is another perspective view of linking gear assembly **158** and pinion gear assembly **164** of actuator **80** (FIGS. **1** and **31**).

(92) As an alternative embodiment, shown in FIG. **35**, push button **82** rests atop center-pin **170**, which extends through upper pinion gear **166**. As can also be seen in FIGS. **35** and **36**, lower pinion gear **168** is engaged with gear rack **106** and includes pinion gear extension **265** that is axially aligned with lower pinion gear **168** that is axially aligned with upper pinion gear **166**. Lower portion **172** of pinion gear **168** extends into opening **174** (FIG. **11**) defined by first locking component housing **150** (FIG. **11**), thereby fixing the position of pinion gear assembly **164** relative to first locking component housing **150** (FIG. **11**), distal bearing **120** (FIG. **11**), first locking component **124** and drive gear **86**, all of which are shown, in a previous embodiment, in FIG. **11**.

(93) FIG. **36** is a perspective view showing engagement of lower pinion gear **168** with gear rack **106** and frustoconical portion **176** of center-pin **170**. As can be seen in FIGS. **37** and **38**, ball bearings **178** extend through side openings **180** defined by pinion gear extension **265** and, when center pin **170** is in an extended position, as shown in FIG. **37**, frustoconical portion **176** of center pin **170** forces ball bearings **178** outwardly and into interfering relation with interference openings **183**, which is defined by side openings **183** and upper pinion gear orifice **182** (FIG. **36**). Upper pinion gear orifice **182** is defined by upper pinion gear **166** (FIGS. **36** and **37**). When interference openings **183** are occupied by ball bearings **178**, upper pinion gear **166** is engaged with lower pinion gear **168**.

(94) As can be seen in the transition from FIG. **40A** to **40B**, when center pin **170** is actuated by depressing button **82** (FIG. **1**), center pin **170** moves within lower pinion gear orifice **263**, and ball bearings **178** are forced inward through side openings **180** of pinion gear **265** by rotation of upper pinion gear **166** relative to lower pinion gear **168** about axis **269**, whereby upper pinion gear **166** is no longer engaged with lower pinion gear **168**. Centerpin **170** is biased in an outward position by biasing spring **184** and frustoconical portion **176**, whereby upper pinion gear **166** is directed into engagement with lower pinion gear **168** by spring **184** located at the base of center pin **170** within pinion gear extension orifice **263** of pinion gear extension **265**. As can be seen at FIGS. **40A** and **40B**, clutch **260** is at lower extension **262**, in which case clutch **260** would be seated in socket **174**, not shown, as explained below. In another embodiment, shown in FIGS. **40B** and **40C**, clutch is at pinion gear **265** above lower pinion gear **164**. In this embodiment, clutch **260** would be seated in a socket (not shown) at an opening of housing **81** through which center pin **170** extends.

(95) FIG. 41 is an exploded view of the embodiment shown in FIGS. 35 through 40A and B, showing one-way needle roller bearing clutch 260 at lower extension 262 of pinion gear assembly 164. Lower extension 262 is located within one-way needle roller bearing clutch 260 which, in turn is seated, such as by being press-fit, into socket 174, shown in FIG. 41 of first locking component housing 150. A side view of pinion gear assembly 164 when assembled with one-way needle roller bearing clutch 260 and first locking mechanism 38, is shown in FIG. 43. During use, the surgeon can depress center pin 170 to compress biasing spring 184 and release ball bearings 178 from outward disposition by releasing frustoconical portion 176 of center pin 170, to thereby disengage lower pinion gear 168 from upper pinion gear 166. By doing so, the surgeon can advance a vascular prosthesis to be delivered by the delivery device of the invention to a position distal to the surgical site without rotating proximal handle 36, whereby lower pinion gear 168 (FIGS. 40-42) will continue to rotate because it continues to be engaged with pinion rack 106 (FIGS. 36-38). However, because upper pinion gear 166 is disengaged from lower pinion gear 168, upper pinion gear 166 will not rotate during longitudinal movement of proximal handle 36 and first locking mechanism 38 along handle body 20. If the surgeon releases proximal handle 36, or pulls handle 36 in a proximal direction (toward the surgeon), lower pinion gear 168 will spin in an opposite direction to that of longitudinal advancement of the vascular prosthesis. Rotation of lower pinion gear 168 in the opposite direction of advancement of vascular prosthesis, will cause one-way needle roller bearing clutch 260 will lock to lock onto lower extension 262, thereby providing resistance to further rotation of lower pinion gear 168 and, consequently, further proximal longitudinal movement of first locking mechanism 38 and the vascular prosthesis toward the surgeon. The resistance to further rotation of the lower pinion gear can be overcome by the surgeon pulling on proximal handle 36 in a proximal direction. Alternatively, proximal handle 36 can be rotated in a counterclockwise direction wherein lower pinion gear extension 262 slips rotationally within the clutch needle rollers so that the resistance to rotation is greater within the interference fit of clutch 260 and socket 174. The resistance to further rotation is friction between at least the one-way needle roller bearing clutch 260 and at least one of socket 174 in which one-way needle roller bearing clutch 260 is press-fit and lower extension 262 of lower pinion gear 168 which is seated in one-way needle roller bearing clutch 260. Upon approach of vascular prosthesis to the surgical site where the vascular prosthesis is to be deployed, the surgeon can release center pin 170 to thereby reengage upper pinion gear 168 with lower pinion gear 166.

(96) Once upper pinion gear 166 is reengaged with lower pinion gear 168, and proximal handle 36 (e.g. FIG. 1) is rotated in a direction, such as a clockwise direction from the surgeon's point of view, the vascular prosthesis can be advanced to a surgical site in a more controlled manner. During advancement of the vascular prosthesis by rotation of proximal handle 36, the vascular prosthesis typically is longitudinally compressed by within delivery device 10. If the surgeon releases proximal handle 36, the vascular prosthesis will exert a proximal longitudinal force (toward the surgeon) on first locking mechanism 36 and, consequently, linking gear assembly 158. Proximal longitudinal force on pinion gear assembly 164 will prompt rotation of lower pinion gear 168 because it is engaged with gear rack 106, thereby causing one-way needle roller bearing clutch 260 to lock onto lower extension 262 of lower pinion gear 168, preventing further rotation in the same direction. Also, lower pinion gear 168 and upper pinion gear 166 are locked because center pin 170 is not actuated, causing ball bearings 178 to be in interfering relation with rotation of upper pinion gear 166, thus preventing backspin of proximal handle 36 in a direction opposite to that of longitudinal advancement of the vascular prosthesis toward the surgical site, and preventing longitudinal expansion of the vascular prosthesis caused by relaxation of longitudinal compression of the vascular prosthesis, such as could be the result of the surgeon releasing the proximal handle 36 after rotation by the surgeon in a direction, such as a clockwise direction to advance the vascular prosthesis to a surgical site. The surgeon can override the friction between one-way needle roller bearing clutch 260 and at least one of socket 174 and lower extension 262 of lower pinion gear 168

by rotating proximal handle **36** in a direction opposite to that which causes advancement, such as a counter clockwise direction. At any time before or after approaching the surgical site, center pin **170** can be depressed to disengage lower pinion gear **168** from upper pinion gear **166**, thereby again allowing the surgeon to longitudinally advance or retract the vascular prosthesis by directing proximal handle **36** distally or proximally without rotation of proximal handle **36**.

(97) As can be seen in FIG. **46A**, nose cone **50** is fixed to guidewire catheter **12** at a distal end **16** of the guidewire catheter **12**. Vascular prosthetic component **58** is disposed within delivery device **10** proximal to nose cone **50**.

(98) FIGS. **44** and **45** show perspective and cut-away views, respectively, of the proximal clasp assembly **184** component of the invention. As can be seen in FIG. **44**, outer coupling **186** is slideable along proximal end **34** of push rod **32**. Fixed component **188** is fixed to the proximal end of the guidewire catheter by pin **192**. Outer coupling **186** and fixed component **188** are in mating relation at juncture **190**. Spring **194** within outer coupling **186** biases outer coupling **186** against fixed component **188**. Proximal clasp assembly **184** is moved from a first position, shown in FIGS. **44**, **47B** to a second position, shown by applying pressure to tongues **196** on either side of outer coupling **186**, and directing outer coupling **186** distally in sufficient degree to allow rotation of outer coupling **186** ninety (90) degrees and then retracting outer coupling **186** so that tongues **196** of outer coupling **186** align between tongues **198** of fixed component **188**, as shown in FIG. **51B**. Movement of outer coupling **186** from the first position, shown in FIG. **44**, to the position shown in FIG. **51B**, causes opening of apex clasp assembly **52**, whereby proximal capture component is retracted from a first position that is in mating relation to the distal capture component **56** of apex clasp assembly **52** shown in FIG. **50B**, to a second position, shown in FIG. **51C**, wherein proximal capture component **54** is no longer in mating relation with distal capture component **56**. Proximal movement of outer coupling **186** of proximal clasp assembly **184** (FIGS. **44**, **47B**, **51B**) relative to a fixed component **188** to separate proximal capture component **54** (FIG. **50B**) from distal capture component **56** (FIGS. **50B**, **51C**) releases apices **68** of stent **66** at proximal end **60** of vascular prosthetic component **58**.

(99) FIGS. **46A-46C** are cross sectional views of a portion of delivery device **10** of the invention showing a vascular prosthetic component **58** in an undeployed state within a distal end **202** of delivery device **10**. Specifically, as shown in FIG. **46A**, vascular prosthetic component **58** is within delivery sheath **200**. Distal end **62** of vascular prosthetic component **58** abuts buttress **204**. Buttress **204**, in turn, is mated to push rod **32** at distal end **206**, proximal end **60** of vascular prosthetic component **58** captured at apices **68** of proximal stent **66** with apex clasp assembly **52** when apex clasp assembly **52** is in a closed position, as shown in FIG. **46A**. Apex clasp assembly **52** includes distal capture component **56** at distal end **16** of guidewire catheter **12**, and proximal capture component **54** is in mateable relation to distal capture component **56**, and attached to distal end **210** of apex release catheter **154**. Apex release catheter **154** extends about guidewire catheter **12**, and both apex release catheter **154** and guidewire catheter **12** extend through vascular prosthetic component **58** and push rod **32** to proximal clasp assembly **184** (FIG. **45**). Delivery sheath **200** is fixed at its proximal end to delivery catheter **28** at distal end **30** and extends about vascular prosthetic component **58** to apex clasp assembly **52**, as can be seen in FIG. **46C**. Returning to FIG. **46A**, nose cone **50** is fixed at guidewire catheter **12** distally to distal capture component **56** of apex clasp assembly **52**. Outer catheter **48** extends from distal handle nose **44** (FIG. **1**), and about delivery catheter **28** and delivery sheath **200**, to nose cone **50**.

(100) As shown in FIGS. **47A-52B**, a method for delivering a vascular prosthesis to a treatment site of the subject employing a delivery device of the invention includes advancing vascular prosthesis **58**, while prosthesis **58** is mounted to apex clasp assembly **52** at proximal end **60** of the prosthesis **58**. Proximal apex clasp assembly **184** is in a first position shown in FIG. **47B**, whereby apex clasp assembly **52** is closed (FIG. **50B**). Apices of vascular prosthesis **58** are secured at apex clasp assembly **52** when proximal clasp assembly **184** is in the first position. Apex clasp assembly **52** is,

in turn, fixed to distal end **16** of guidewire catheter **12**, shifting knob **42** is in a first position when pin **108** is in slot **110** (FIG. 47C), causing push rod **32** to move with longitudinal movement of proximal handle **36**. Prosthesis **58** is advanced to a position distal to a vascular treatment site of the subject by rotation of proximal handle **36** in a first direction about handle body **20**, having distal end **26**, of delivery device **10** through which guidewire catheter **12** extends. Guidewire catheter **12** is disposed within push rod **32** that also extends through handle body **20**, wherein guidewire catheter **12** is fixed to push rod **32**, such as at a proximal end of guidewire catheter **12** or push rod **32** by pin **192** (FIG. 44), whereby rotation of proximal handle **36** causes longitudinal movement of guidewire catheter **12** and push rod **32** along handle body **20** to thereby at least partially advance prosthesis **58** from outer catheter **48** as can be seen in FIGS. 48A-48B. Optionally, push button **82** of actuator **80** can be depressed to disengage rotation of proximal handle **36** from longitudinal movement of proximal handle **36** along handle body **20**, to thereby allow manual advancement of vascular prosthesis **58** to the vascular treatment site of the subject without rotation of proximal handle **36** about handle body **20**.

(101) Shifting knob **42** is shifted from a first position, wherein first locking component **124** (FIGS. 10, 11) secures proximal handle **36** to push rod **32**, to a second position, whereby first locking component **124** (FIGS. 10, 11) disengages proximal handle **36** from push rod **32** and second locking component **144** (FIGS. 10, 11) engages push rod **32** with handle body **20** at proximal end **24** of handle body **20**.

(102) As can be seen in FIGS. 50A and 50B, proximal handle **36** can then be rotated in a second direction, while actuator push button **82** is not depressed, whereby delivery catheter **28**, having a distal end **30** (FIG. 53A) and extending about push rod **32**, is withdrawn along push rod **32**, and delivery sheath **200** extending from distal end of the delivery catheter (FIGS. 4 through 9) is at least partially retracted from about prosthesis **52**. Optionally, push-button **82** of actuator **80** can be depressed, thereby disengaging rotation of proximal handle **36** from handle body **20**, to thereby fully retract of delivery sheath **200** from vascular prosthesis **58** without rotation of proximal handle **36** about handle body **20**, as can be seen in FIG. 51A.

(103) Proximal clasp assembly **184** is then actuated by compressing outer coupling **186** and moving outer coupling **186** first distally, then rotating outer coupling **186** ninety degrees, and thereafter retracting outer coupling **186** to a second position, shown in FIG. 51B, thereby retracting apex release catheter **154** within push rod **32** (FIGS. 10, 11) and retracting proximal capture component **54** from distal capture component **56**. Apices **68** of stent **66** at the proximal end **60** of vascular prosthesis **58** are released from apex clasp assembly **52**, and prosthesis **58** is thereby released from the delivery device **10**, as can be seen in FIG. 51C. Shifting knob **42** is then moved from the second position to the third position, wherein pin **108** is located in slot **114** between first slot **110** and second slot **112**, as can be seen in FIG. 52B, thereby disengaging push rod **32** from handle body **20**. Push rod **32** and guidewire catheter **12** are then withdrawn from vascular prosthesis **58** by pulling push rod **32** through handle body **20**, thereby completing delivery of vascular prosthesis **58** to the treatment site, as can be seen in FIG. 52A.

(104) While this invention has been particularly shown and described with references to example embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the invention encompassed by the appended claims.

(105) The relevant portion of all references cited herein and U.S. Pat. Nos. 8,070,790, 9,101,506, 9,364,314, 9,554,929, and 10,299,951, and U.S. patent application Ser. No. 16/413,916 (Publication No.: 2019/0269539), are incorporated by reference in their entirety.

Claims

1. A delivery device (10), comprising: a) a handle body (20), having a longitudinal axis (22), a proximal end (24), and a distal end (26); b) a gear rack (106) extending within the handle body (20); c) a proximal handle (36) extending about the gear rack (106) and defining teeth (156), the proximal handle (36) being rotatable about the handle body (20) and the gear rack (106); d) a distal handle (40) that extends around the handle body (20) at the distal end (26) of the handle body (20); e) a guidewire catheter (12) having a proximal end (14) and a distal end (16), the guidewire catheter (12) extending through the handle body (20), the proximal handle (36), and the distal handle (40), and along the longitudinal axis (22); f) a delivery catheter (28) axially fixed to the proximal handle (36), and having a distal end (30) extending from within the distal end (26) of the handle body (20) and about the guidewire catheter (12); g) an outer catheter (48) extending distally from the distal handle (40), and about the delivery catheter (28) when the delivery catheter (28) is in a first, retracted position; h) a gear assembly linking the teeth (156) of the proximal handle (36) to the gear rack (106), whereby rotation of the proximal handle (36) about the longitudinal axis (22) moves the proximal handle (36) and the delivery catheter (28) along the longitudinal axis (22) relative to the gear rack (106); i.) the gear assembly further including: 1) A pinion gear assembly (164) engaging the gear rack (106); and 2) a linking gear assembly (158) including an upper linking gear (160) engaging the teeth (156) of the proximal handle (36) and a lower linking gear (162) fixed to the upper linking gear (160) and between the upper linking gear (160) and the longitudinal axis (22) of the handle body (20), and having teeth engaging the pinion gear assembly (164), the upper linking gear (166) and the lower linking gear (162) having a common axis of rotation that is normal to the longitudinal axis (22) of the handle body (20), wherein the upper linking gear (160) and the lower linking gear (162) each define a central opening along the common axis of rotation; j) a clutch at the gear assembly, whereby the clutch engages the gear assembly with the proximal handle and thereby biases longitudinal movement of the gear assembly and associated rotation of the proximal handle (36) about the longitudinal axis; wherein rotation of the proximal handle (36) about the longitudinal axis (22) when the clutch is engaged is resisted by friction between the clutch and at least one of the gear assembly and a portion of the remainder of the delivery device (10), and wherein the clutch engages when the gear assembly is directed in a proximal direction along the longitudinal axis; k) a push rod (32) extending about the guidewire catheter (12) and within the delivery catheter (28), the push rod (32) being fixed to the guidewire catheter (12) at the proximal end of the guidewire catheter (12) proximal to the handle body (20), and selectively fixed to the proximal handle (36); and l) a locking mechanism assembly extending about the push rod, the locking mechanism including i.) a first locking mechanism (38) that locks the delivery catheter to the push rod when the locking mechanism is in a first locking position, and ii.) a second locking mechanism (132) fixed to the distal end (26) of the handle body (20) that locks the push rod (32) to the handle body (20) when the locking mechanism assembly is in a second locking position, wherein the first locking position and the second locking position are mutually exclusive m.) An actuator (80) that includes the gear assembly and which further includes i.) a housing (81) extending about the handle body (20) and having a proximal end defining a proximal opening and a distal end defining a distal opening, the housing (81) further defining an aperture between the proximal opening and the distal opening; and ii.) a center pin extending through the central openings of the upper linking gear (160) and the lower linking gear (162), the pin including the center pin including opposite ends that are at the locking mechanism assembly and the housing (81), wherein the clutch selectively engages the pin with the linking gear assembly whereby engagement depends upon the direction of rotation of the proximal handle (36) about the handle body (20); n) wherein the linking gear assembly (158) further includes a second clutch that engages the linking gear assembly (158) with the pin when the pinion gear assembly (164) is directed distally along the gear rack (106) during the engagement of the linking gear assembly (158) with the pinion gear assembly (164); o) wherein the second clutch is a second one-way

needle roller clutch bearing that is press-fit into a first socket or a second socket into which the pin extends, whereby the second one-way needle roller clutch bearing engages the linking gear assembly (158) with the pin when the pinion gear assembly (163) is directed proximally along the gear rack (106) during engagement of the linking gear assembly (158) with the pinion gear assembly (164), and wherein resistance to proximal movement of the pinion gear assembly while the second one-way needle roller clutch is engaged is caused by at least one of resistance to rotation of the second roller needle clutch bearing within the socket within which the one-way needle roller bearing is press-fit or the interface between the pin and the second one-way needle roller clutch, and wherein the resistance is less than that between the first roller needle clutch and the pin.

2. The delivery device of claim 1, wherein the gear assembly includes a pinion gear assembly (164) that includes a) an upper pinion gear (166) engaged with the proximal handle (36), the upper pinion gear (166) defining a non-circular pinion gear orifice (182) and rotatable about a pinion gear axis (269); and b) a lower pinion gear (168) axially aligned with the upper pinion gear (166) and defining a lower pinion gear orifice (263), the lower pinion gear (168) engaged with the gear rack (106) and selectively engaged with the upper pinion gear (166), wherein the clutch (260) engages the lower pinion gear (168) when the gear assembly is directed in a proximal direction.

3. The delivery device of claim 2, wherein the lower pinion gear (168) includes a. a lower portion extending toward the longitudinal axis (22) of the handle body (20); b. a gear portion engaged with the gear rack (106); c. a pinion gear extension (265) that extends within the upper pinion gear orifice (182), wherein the pinion gear extension (265) defines a side opening (180), the side opening (180) and the upper pinion gear orifice (182) of the upper pinion gear (166) together defining an interference opening (183) that, when occupied, prevents rotation of the upper pinion gear (166) and the lower pinion gear (168) relative to each other; d. a ball bearing (178) in at least one of each side opening (180), the ball bearing (178) having a diameter greater than a thickness of a wall defining the side opening (180); e. a center pin (170) moveable along the pinion gear axis and within the upper pinion gear orifice, the lower pinion gear orifice, and the pinion gear extension orifice, the center pin (170) including a frustoconical portion (176) between a base portion having a first diameter within the lower pinion gear orifice and a second diameter that is less than the first diameter, and located in the upper pinion gear orifice, whereby movement of the frustoconical portion (176) of the center pin (170) causes radially outward displacement of the ball bearing (178) into the interference opening (183), thereby causing an interfering relation between rotation of the upper pinion gear (166) and the lower pinion gear (168); and f. a spring (184) at the lower pinion gear (168) that provides bias to the center pin (170) radially outward from the longitudinal axis (22) of the handle body (20), whereby the ball bearing (178) is directed radially outward through the side opening (180), thereby causing the interfering relation between the upper pinion gear (166) and the lower pinion gear (168), whereby depressing the center pin (170) removes outward displacement of the ball bearing (178) and eliminates the interfering relation between rotation of the upper pinion gear (166) and the lower pinion gear (168) to cause rotation of the proximal handle (36) to be independent of longitudinal movement of the delivery catheter (28) relative to the handle body (20) along the longitudinal axis (22).

4. The delivery device of claim 1, wherein the pin is integral to or fused with at least one of the locking mechanism assembly and the housing (81).

5. The delivery device of claim 1, wherein the first locking mechanism (38) defines a first socket (174) and the housing (81) defines a second socket, and wherein the opposite ends of the pin are seated in the first socket (174) and the second socket.

6. The delivery device of claim 5, wherein the clutch engages the pin when the gear assembly is directed in a proximal direction.

7. The delivery device of claim 6, wherein rotation of the proximal handle (20) that directs the gear assembly in a proximal direction is resisted by an interfering relationship between the pin and at

least one of the first and the second sockets.

8. The delivery device of claim 6, wherein rotation of the proximal handle (20) that directs the gear assembly in a proximal direction is resisted by an interfering relationship between the clutch and at least one of the first and the second sockets.

9. The delivery device of claim 6, wherein rotation of the proximal handle (20) that directs the gear assembly in a proximal direction is resisted by an interfering relationship between the clutch and the pin.

10. The delivery device of claim 1, wherein the second clutch is a coil spring that is fixed at one end to the first locking mechanism (38) and the pin extends through the coil spring, whereby the coil spring is engaged with the pin when the pinion gear assembly (164) is directed distally along the gear rack (106) during engagement of the linking gear assembly (158) with the pinion gear assembly (164).

11. The delivery device of claim 1, wherein the second clutch is a coil spring that is fixed at one end to the housing (81) and the pin extends through the coil spring, whereby the coil spring is engaged with the pin when the pinion gear assembly (164) is directed distally along the gear rack (106) during engagement of the linking gear assembly (158) with the pinion gear assembly (164).

12. A delivery device (10), comprising: a. a handle body (20), having a longitudinal axis (22), a proximal end (24), and a distal end (26); b. a gear rack (106) extending within the handle body (20); c. a proximal handle (36) extending about the gear rack (106) and defining teeth (156), the proximal handle (36) being rotatable about the handle body (20) and the gear rack (106); d. a distal handle (40) that extends around the handle body (20) at the distal end (26) of the handle body (20); e. a guidewire catheter (12) having a proximal end (14) and a distal end (16), the guidewire catheter (12) extending through the handle body (20), the proximal handle (36), the distal handle (40), and along the longitudinal axis (22); f. a delivery catheter (28) axially fixed to the proximal handle (36), and having a distal end (30) extending from within the distal end (26) of the handle body (20) and about the guidewire catheter (12); g. an outer catheter (48) extending distally from the distal handle (40), and about the delivery catheter (28) in a first, retracted position; h. a gear assembly linking the teeth (156) of the proximal handle (36) to the gear rack (106), whereby rotation of the proximal handle (36) about the longitudinal axis (22) moves the proximal handle (36) and the delivery catheter (28) along the longitudinal axis (22) relative to the gear rack (106), wherein the gear assembly includes i. A pinion gear assembly (164) engaging the gear rack (106), and ii. a linking gear assembly (158) including an upper linking gear (160) engaging the teeth (156) of the proximal handle (36) and a lower linking gear (162) fixed to the upper linking gear (160) and between the upper linking gear (160) and the longitudinal axis (22) of the handle body (20), and having teeth engaging the pinion gear assembly (164), the upper linking gear (166) and the lower linking gear (162) having a common axis of rotation that is normal to the longitudinal axis (22) of the handle body (20), wherein the upper linking gear (160) and the lower linking gear (162) each define a central opening along the common axis of rotation; i. a push rod (32) extending about the guidewire catheter (12) and within the delivery catheter (28), the push rod (32) being fixed to the guidewire catheter (12) at the proximal end of the guidewire catheter (12) proximal to the handle body (20), and selectively fixed to the proximal handle (36); j. a locking mechanism assembly extending about the push rod, the locking mechanism including i. a first locking mechanism (38) that locks the delivery catheter to the push rod when the locking mechanism is in a first locking position, wherein the first locking mechanism (38) defines a first socket (174) and the housing (81) defines a second socket, and wherein the opposite ends of the pin are seated in the first socket (174) and the second socket, and ii. a second locking mechanism (132) fixed to the distal end (26) of the handle body (20) that locks the push rod (32) to the handle body (20) when the locking mechanism assembly is in a second locking position, wherein the first locking position and the second locking position are mutually exclusive; k. an actuator (80) that includes the gear assembly and which further includes i. a housing (81) extending about the handle body (20) and having a proximal end

defining a proximal opening and a distal end defining a distal opening, the housing (81) further defining an aperture between the proximal opening and the distal opening, and ii. a center pin extending through the central openings of the upper linking gear (160) and the lower linking gear (162), the pin including opposite ends that are at the locking mechanism assembly and the housing (81); and l. A clutch at the gear assembly, whereby the clutch engages the gear assembly with the proximal handle and thereby biases longitudinal movement of the gear assembly and associated rotation of the proximal handle (36) about the longitudinal axis, wherein rotation of the proximal handle (36) about the longitudinal axis (22) when the clutch is engaged is resisted by friction between the clutch and at least one of the gear assembly and a portion of the remainder of the delivery device (10), wherein the clutch engages the pin with the linking gear assembly when the gear assembly is directed in a proximal direction, and wherein rotation of the proximal handle (20) that directs the gear assembly in a proximal direction is resisted by an interfering relationship between the pin and at least one of the first and the second sockets; m. wherein the linking gear assembly (158) further includes a second clutch that engages the linking gear assembly (158) with the pin when the pinion gear assembly (164) is directed distally along the gear rack (106) during the engagement of the linking gear assembly (158) with the pinion gear assembly (164), wherein the second clutch is a second one-way needle roller clutch bearing that is press-fit into the first socket or the second socket into which the pin extends, whereby the second one-way needle roller clutch bearing engages the linking gear assembly (158) with the pin when the pinion gear assembly (163) is directed proximally along the gear rack (106) during engagement of the linking gear assembly (158) with the pinion gear assembly (164), and wherein resistance to proximal movement of the pinion gear assembly while the second one-way needle roller clutch is engaged is caused by at least one of resistance to rotation of the second roller needle clutch bearing within the socket within which the one-way needle roller bearing is press-fit or the interface between the pin and the second one-way needle roller clutch, and wherein the resistance is less than that between the first roller needle clutch and the pin.
